HACKHUSTLE 2.0

USER-CONTROLLES FINANCIAL IDENTITY VERIFICATION

Design and develop a user-controlled financial identity verification system that addresses deep-fake-powered impersonation, synthetic identity fraud, and credential theft in regulated financial services.

PROBLEM STATEMENT NUMBER	-	001
PROBLEM STATEMENT TITLE	-	User-controlled financial identity verification system

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CERTIFICATE

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ACKNOWLEDGEMENT

I would like to express my heartfelt gratitude to our esteemed Founder President **Dr. N. M. Veeraiyan**, President **Dr. Saveetha Rajesh**, Director **Dr. S. Rajesh**, and the entire management team for providing the essential infrastructure.

I extend my sincere appreciation to the principal, **Dr. V. Vijaya Chamundeeswari, M.Tech., Ph.D.,** for creating a supportive learning environment for this project.

I am very thankful to the Dean of ICT, **Prof. Obed Otto, M.E.,** for facilitating a conducive atmosphere that helped me to complete my project successfully.

I am also thankful **to Dr. G. Nagappan**, Professor and Associate Dean of ICT at Saveetha Engineering College, for his generous support and for providing the necessary resources for the mini project work.

I would also like to express my profound gratitude to my Supervisor, **Dr. R. N. Karthika**, Assistant Professor at Saveetha Engineering College, for their invaluable guidance, suggestions, and constant encouragement, which were instrumental in the successful completion of the mini project. Their timely support and insights during the review process were greatly appreciated.

I am grateful to all faculty, staff, and technicians for their cooperation throughout the mini project. Finally, I wish to acknowledge my loving parents, friends, and well-wishers for their encouragement in helping to achieve this milestone.

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ABSTRACT

The rapid digitization of financial services has introduced significant vulnerabilities in identity verification systems, particularly due to the rise of deepfake-based impersonation, synthetic identity fraud, and credential theft. Traditional authentication mechanisms, which rely on static credentials or single-layer biometric verification, are increasingly insufficient in preventing sophisticated attacks. The lack of real-time identity validation, combined with fragmented identity storage across institutions, creates critical security gaps and reduces user control over personal data. The proposed Portable Verified Identity Financial Wallet (TrustVault) introduces a next-generation identity verification framework that enables secure, user-controlled, and real-time authentication. Unlike conventional systems, TrustVault integrates live face recognition, liveness detection, deepfake detection, and behavioral verification into a unified multi-layer architecture. Each authentication attempt is validated dynamically, ensuring that the user is physically present and not a synthetic or replayed entity. The system is designed with a modular and privacy-centric architecture, incorporating cryptographic hashing, decentralized storage, and optional blockchain-based audit trails. Identity credentials are not merely stored but are actively verified during every transaction, ensuring continuous trust validation. This approach transforms identity verification from a static checkpoint into a real-time, intelligent, and tamper-resistant process. By embedding security at multiple layers and providing users with control over their identity data, TrustVault enhances transparency, reduces fraud risk, and establishes a scalable framework for secure financial authentication across banking, digital payments, and regulatory environments.

Keywords:

Financial Identity Verification, Deepfake Detection, Liveness Detection, Behavioral Authentication, Portable Identity Wallet, Real-Time Verification, Secure Authentication, Decentralized Identity, Biometric Security

INTRODUCTION

In modern financial ecosystems, identity verification is a foundational requirement for enabling secure transactions, regulatory compliance, and user trust. With the expansion of digital banking, fintech platforms, and online financial services, users are required to authenticate themselves across multiple systems and institutions. However, existing identity verification frameworks are largely fragmented, relying on static credentials, one-time verification processes, and centralized databases that lack real-time validation capabilities. As organizations scale, identity-related vulnerabilities become more pronounced. Attackers increasingly exploit advanced technologies such as deepfake generation, synthetic identity creation, and credential injection attacks to bypass traditional verification mechanisms. Static biometric systems, such as stored facial templates, are particularly vulnerable to replay attacks and AI-generated impersonation. Furthermore, the absence of continuous verification allows unauthorized users to gain access after initial authentication. Another critical limitation lies in the lack of user control over identity data. In conventional systems, user credentials are stored across multiple platforms, increasing the risk of data breaches and unauthorized access. Identity ownership is effectively transferred to institutions, reducing transparency and accountability. The problem is further exacerbated by the absence of integrated, real-time identity validation mechanisms. Financial institutions often rely on delayed verification processes, which detect fraud only after it has occurred. This reactive model leads to financial losses, regulatory challenges, and reduced trust in digital systems.

To address these challenges, the proposed system introduces a Portable Verified Identity Financial Wallet, which enables users to securely manage and verify their identity in real time. The system integrates live biometric verification, AI-driven fraud detection, and cryptographic security mechanisms to ensure that identity authentication is dynamic, tamper-proof, and user-controlled.

LITERATURE REVIEW

Modern identity verification systems operate within increasingly complex digital environments where authentication processes span multiple platforms, devices, and regulatory frameworks. Traditional identity management approaches rely on centralized databases and static verification mechanisms, which introduce vulnerabilities related to data breaches, identity duplication, and delayed fraud detection. These systems often validate identity only at the initial point of access, without continuous verification during subsequent interactions.

Biometric authentication has been widely adopted as a solution to enhance security. Technologies such as facial recognition, fingerprint scanning, and iris detection provide a higher level of assurance compared to password-based systems. Models like FaceNet have demonstrated high accuracy in facial embedding and matching. However, these systems remain susceptible to spoofing attacks, including replay attacks using recorded videos and AI-generated deepfakes.

Recent research highlights the growing threat of deepfake technologies, which leverage Generative Adversarial Networks (GANs) to create highly realistic synthetic identities. These attacks can bypass traditional biometric systems by mimicking facial features and expressions. Studies emphasize the need for multi-layer verification mechanisms that incorporate liveness detection and anomaly analysis to distinguish real users from synthetic entities.

To enhance trust and transparency, decentralized identity frameworks have been proposed. Blockchain-based systems provide immutable audit trails and tamper-resistant storage, reducing reliance on centralized authorities. Platforms such as Ethereum enable secure identity verification through smart contracts and cryptographic validation. However, challenges related to scalability, latency, and privacy remain significant concerns.

Behavioral authentication has emerged as an additional layer of security, analyzing user interaction patterns such as typing speed, gesture dynamics, and response timing. Unlike static biometrics, behavioral patterns are difficult to replicate, making them effective in detecting unauthorized access.

Table 1.1 Comparative Analysis of Identity Verification Models

Parameter	Traditional Identity Systems	Portable Verified Identity Wallet (Proactive Authentication)
Verification Approach	Static, one-time authentication	Dynamic, multi-layer real-time verification
Authentication Timing	Only at login stage	Continuous, at every access request
Biometric Usage	Single layer (face/fingerprint)	Multi-layer (face + liveness + deepfake detection)
Liveness Detection	Limited or absent	Integrated real-time liveness verification
Deepfake Protection	Not supported	AI-based deepfake detection
Behavioral Analysis	Not considered	Integrated behavioral pattern analysis
Data Storage	Centralized databases	Secure hashing with optional decentralized storage
Audit Trail	Editable and centralized logs	Tamper-resistant and cryptographically secured logs

Despite these advancements, existing systems often implement these techniques in isolation. The lack of an integrated framework that combines biometric verification, deepfake detection, behavioral analysis, and secure identity storage limits the effectiveness of current solutions.

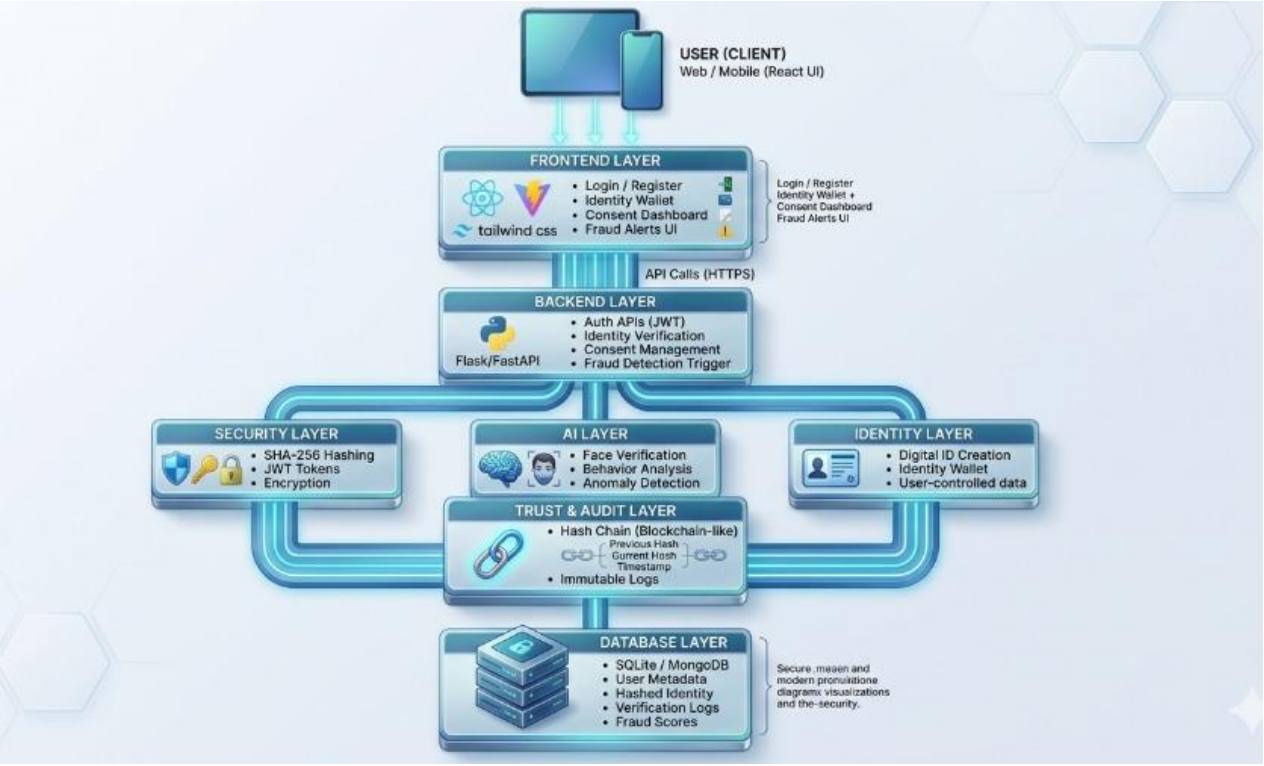
The proposed TrustVault system addresses these limitations by integrating multiple verification layers into a unified architecture, enabling real-time, secure, and user-controlled identity authentication.

DESIGN METHODOLOGY

Portable Identity Verification Framework

The proposed system introduces a Portable Verified Identity Wallet Architecture designed to ensure secure, real-time, and user-controlled

Fig 1.1 Proposed Intelligence Map



The architecture consists of the following core components:

- Live Face Capture Module
- Liveness Detection Engine
- Deepfake Detection Model
- Behavioral Analysis Engine
- Identity Matching System
- Secure Storage Layer (Hash/Blockchain)

The system ensures that every authentication request is validated through multiple layers before granting access.

ALGORITHM:

INPUT: U, F, B, L, D, Hflag
OUTPUT: VID, AuthStatus, TxLog

```
function PVIFW(U, F, B, L, D, Hflag)
  begin
    Capture ← LiveFace(F)
    Liveness ← CheckLiveness(L)
    if Liveness == False then
      return Error: "Spoof Detected"
    end if

    Deepfake ← DetectDeepfake(D)
    if Deepfake == True then
      return Error: "Deepfake Detected"
    end if

    Behavior ← AnalyzeBehavior(B)
    Match ← CompareFace(U, Capture)

    if Match == True AND Behavior == Valid then
      VID ← GenerateHash(U)
      TxLog ← Log(AuthSuccess, U)
      return VID, AuthSuccess, TxLog
    else
      TxLog ← Log(AuthFail, U)
      return Error: "Authentication Failed"
    end if
end
```

As shown in the algorithm, the proposed Portable Verified Identity Financial Wallet integrates live face recognition, liveness detection, deepfake detection, behavioural analysis, and secure hashing into a unified real-time authentication workflow. Initially, the system captures the user's live facial input and performs liveness checks to ensure physical presence. The captured data is then analysed using deepfake detection models to identify any synthetic or manipulated inputs.

Subsequently, behavioural patterns such as response timing and interaction dynamics are evaluated to provide an additional layer of verification. The validated biometric data is compared with stored identity embeddings, and upon successful matching, a secure hash-based identity token is generated. All authentication events are logged to ensure traceability and accountability.

This architecture enables dynamic, multi-layer identity verification, ensuring secure authentication, resistance to spoofing attacks, and user-controlled identity management suitable for modern financial systems.

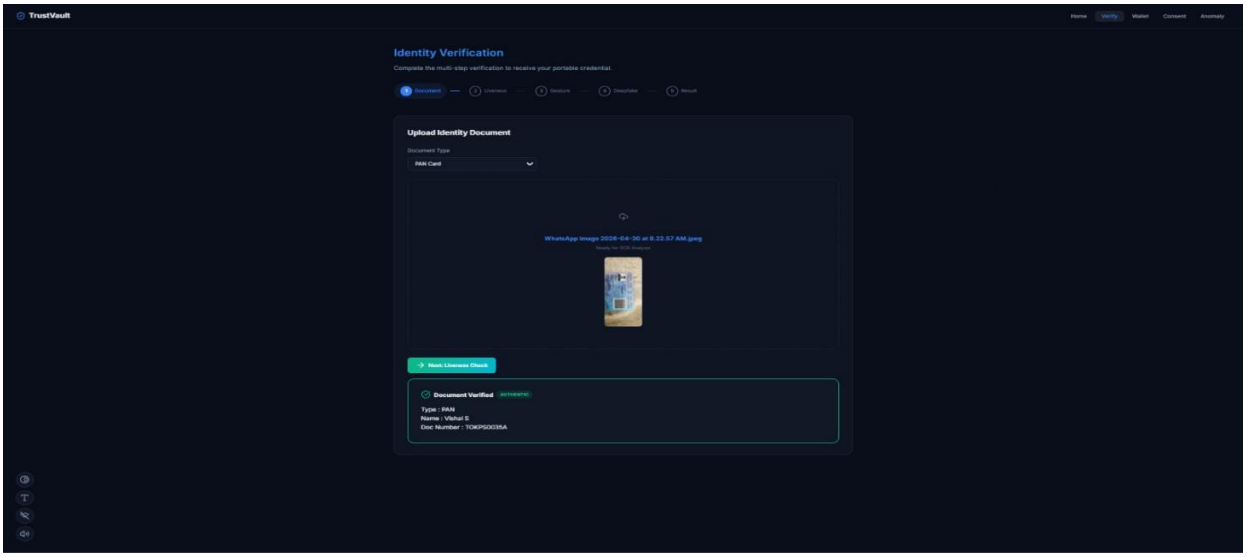
PROTOTYPE TESTING

The developed prototype was tested across all functional modules to validate system behavior, policy enforcement, security controls, and data integrity mechanisms. The following test cases were executed to ensure the system meets the defined objectives.

2.1 Identity Document Upload

Test case ID	**TC_DOC_01**
Module	Identity Enrollment
Objective	To verify that user identity documents (Aadhaar, PAN, etc.) are uploaded and stored securely
Pre-condition	System is active and user identity session initiated
Test Input	Upload valid identity document file (image/PDF)
Expected Output	Document uploaded, validated, and stored securely
Actual Output	Document successfully uploaded and stored

Fig 2.1 Identity Document verification



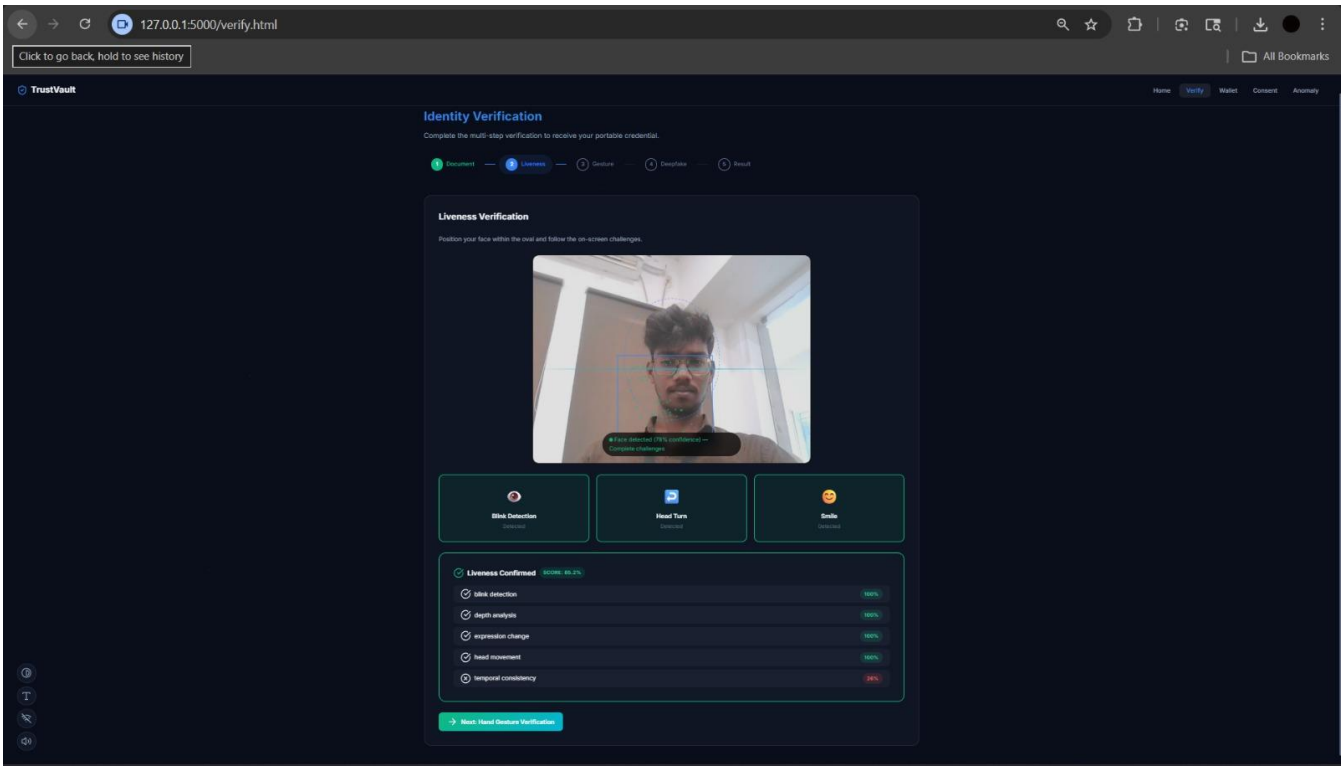
2.2 Live Face Detection

Test Case ID	**TC_FACE_01**
Module	Face Detection
Objective	To verify real-time face detection during authentication
Pre-condition	Camera access enabled
Test Input	User presents face to camera
Expected Output	Face detected successfully
Actual Output	Face detected in real-time

2.3 Liveness Detection

Test Case ID	**TC_LIVE_01**
Module	Liveness Detection
Objective	To ensure system detects real human presence
Pre-condition	Face detected
Test Input	User performs actions (blink/smile/head movement)
Expected Output	System confirms liveness
Actual Output	Liveness verified successfully

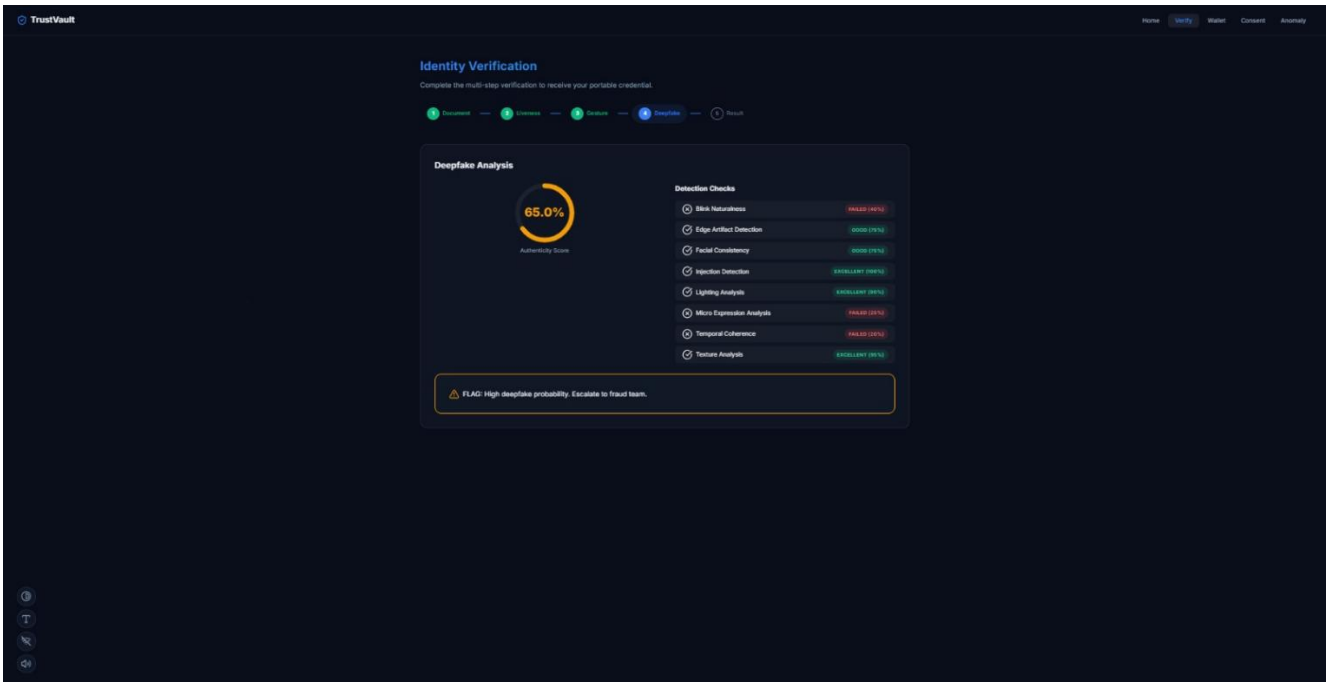
Fig 2.2 Live Face and Liveness Detection



2.4 Deepfake Detection

Test Case ID	**TC_DF_01**
Module	Deepfake Detection
Objective	To detect AI-generated or spoofed facial inputs
Pre-condition	Face input available
Test Input	Synthetic/deepfake image or video
Expected Output	System flags input as fake
Actual Output	Deepfake detected successfully

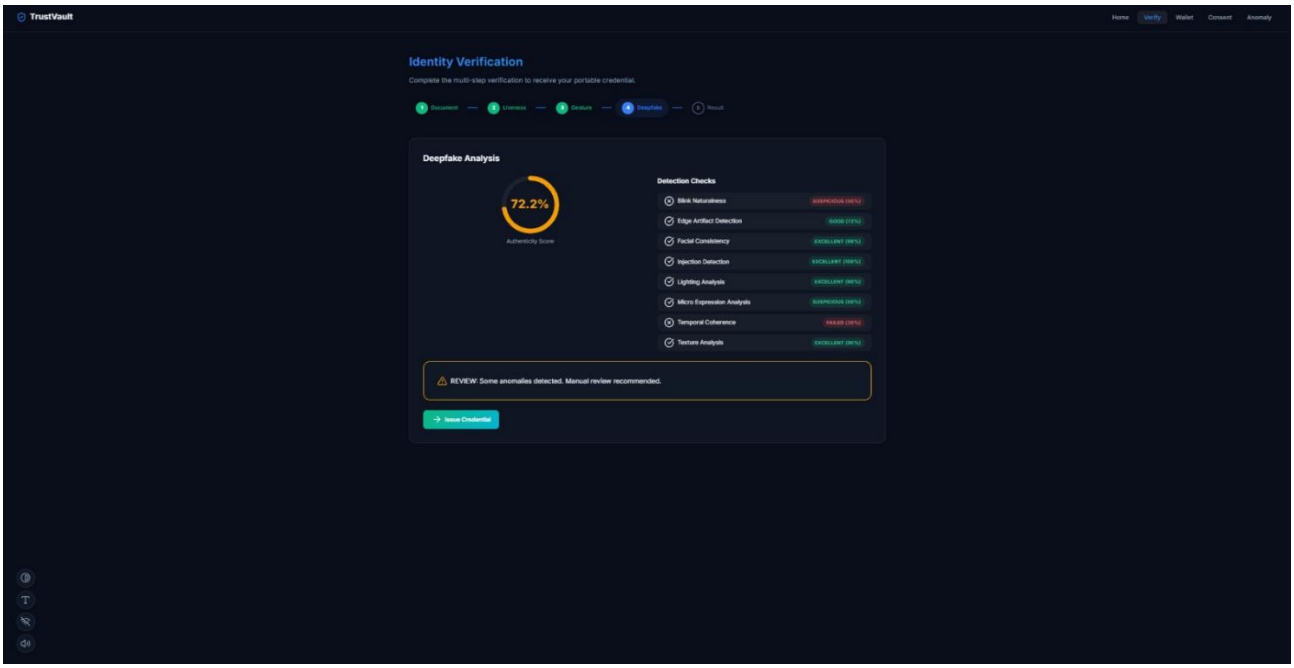
Fig 2.3 Deepfake Detection



2.5 Identity Matching

Test case ID	**TC_MATCH_01**
Module	Identity Verification
Objective	To verify matching of live face with stored identity
Pre-condition	Identity enrolled in system
Test Input	Live face input
Expected Output	Match found for valid identity
Actual Output	Identity matched successfully

Fig 2.4 Identity Matching



2.6 Tampering Detection

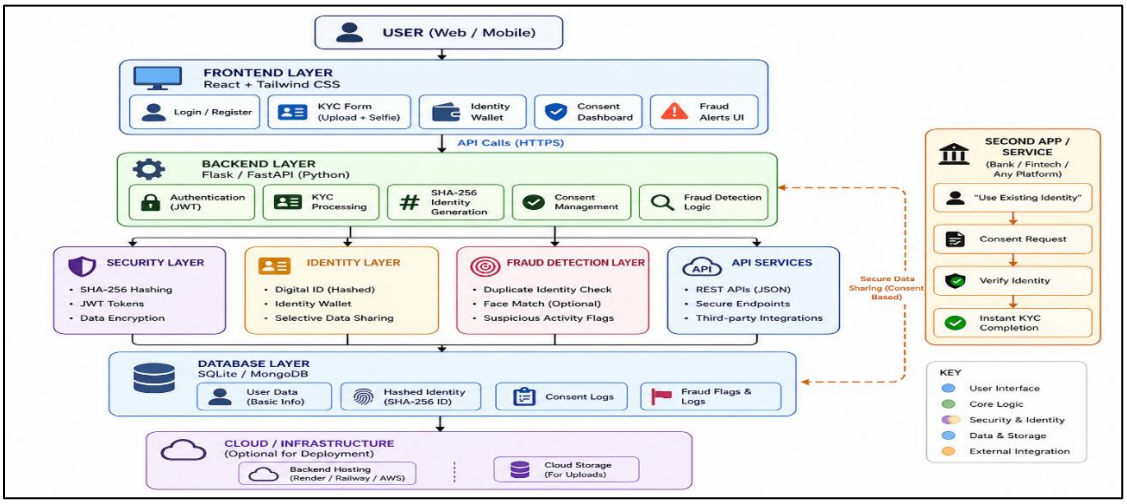
Test case ID	**TC_TAMP_01**
Module	Data Integrity
Objective	To detect unauthorized modification of identity data
Pre-condition	Identity data stored with hash
Test Input	Attempt to alter stored identity data
Expected Output	Hash mismatch detected
Actual Output	Tampering identified successfully

WORKFLOW:

To evaluate the proposed prototype, structured test cases and RESTful API testing using Postman were conducted to ensure correct system functionality, robust security enforcement, and consistency of identity data across all modules. Core functionalities such as user registration, identity enrollment, live face capture, liveness verification, deepfake detection, behavioral analysis, authentication validation, and secure logging were tested to confirm real-time operation and reliability. The APIs are designed in a modular architecture, enabling clear separation between authentication, identity verification, behavioral analysis, and audit management layers. This division of concerns ensures scalability and maintainability while allowing independent enhancement of each component. Multi-layer verification mechanisms, including liveness detection and deepfake analysis, effectively prevent spoofing and unauthorized access, while role-based controls restrict sensitive operations to authorized entities. Secure hashing techniques are implemented to protect identity data and detect any tampering attempts, ensuring data integrity throughout the system. Additionally, all authentication events are recorded in structured logs to maintain traceability and accountability. While the current implementation focuses on centralized secure storage, future integration of blockchain is planned to enhance audit transparency and provide immutable verification records.

The backend workflow demonstrates a seamless data flow between live capture, verification engines, identity matching, and logging components, forming a layered and secure authentication pipeline. Overall, the prototype establishes a reliable, real-time identity verification framework with a scalable architecture, capable of adapting to advanced security requirements in financial systems.

Fig 3.1 Technology Stack



RESULTS AND DISCUSSIONS

In today's digital financial ecosystem, identity verification often begins with a single checkpoint—logging in with credentials or performing a one-time biometric scan. Once access is granted, the system assumes continued legitimacy, leaving a critical gap where unauthorized users can operate undetected. As deepfake technologies and synthetic identity attacks evolve, this static and trust-based model becomes increasingly fragile, allowing sophisticated impersonation techniques to bypass traditional safeguards. Consider a scenario where an attacker uses an AI-generated face or a recorded video to mimic a legitimate user. In conventional systems, such attempts may pass initial verification, especially when liveness detection is weak or absent. The system, having no mechanism for continuous or layered validation, proceeds to grant access, potentially leading to fraudulent transactions and compromised financial data. Detection, if it occurs, is often delayed—only identified during audits or anomaly reviews—by which time the damage has already been done. The proposed Portable Verified Identity Financial Wallet transforms this reactive model into a proactive and intelligent authentication framework. Instead of relying on a single verification step, the system initiates a real-time, multi-layer validation process each time authentication is requested. It begins by capturing a live facial input and immediately performing liveness checks, such as detecting eye movement or guided user actions, to confirm physical presence. Simultaneously, deepfake detection models analyze subtle inconsistencies in facial patterns to identify synthetic or manipulated inputs. Beyond biometrics, the system observes behavioral signals—such as response timing and interaction patterns—to establish an additional layer of identity assurance. These combined inputs are then matched against securely stored identity embeddings. Only when all layers align does the system generate a secure identity token, ensuring that access is granted based on dynamic, verified trust rather than static assumptions. The architecture is designed to scale across diverse financial platforms while maintaining efficiency and adaptability. By modularizing verification components, the system allows continuous improvement of detection models without disrupting core operations. Secure hashing mechanisms and optional decentralized logging ensure that every authentication event is traceable, tamper-resistant, and transparent to authorized stakeholders. In this way, the system not only prevents unauthorized access before it occurs but also establishes a continuous trust framework. It shifts identity verification from a one-time gateway into an ongoing, intelligent

process—significantly reducing fraud risk, enhancing user control, and strengthening confidence in digital financial interactions.

CONCLUSION

The proposed Portable Verified Identity Financial Wallet aims to address a critical gap in modern digital financial systems: the lack of real-time, secure, and user-controlled identity verification in the presence of advanced threats such as deepfakes, synthetic identities, and credential compromise. By tightly integrating multi-layer authentication mechanisms—including live face recognition, liveness detection, deepfake analysis, and behavioral verification—into the identity validation process, the system shifts authentication from a static, one-time checkpoint into a continuous and proactive trust framework. Unlike traditional centralized systems that rely on initial verification and detect anomalies only after unauthorized access has occurred, the proposed solution enforces identity validation at the exact moment of interaction. Each authentication request is dynamically evaluated, ensuring that only genuine users are granted access before any financial action takes place. The incorporation of secure hashing mechanisms, tamper-resistant logging, and optional decentralized audit trails ensures that every authentication event is transparent, verifiable, and resistant to manipulation. Furthermore, the modular architecture of the system enhances scalability, flexibility, and maintainability, allowing it to adapt seamlessly to evolving security threats and complex financial environments. By decoupling identity verification components, the platform supports continuous improvement in detection models without disrupting core operations. The privacy-centric design ensures that sensitive identity data remains protected while still enabling controlled visibility for regulatory and institutional requirements. The integration of real-time verification, intelligent fraud detection, and user-centric identity control significantly reduces the risk of unauthorized access, minimizes identity fraud, and enhances trust in digital financial interactions. Ultimately, the proposed system establishes a robust, future-ready framework for secure identity management, empowering financial institutions and users to operate with confidence, transparency, and resilience in increasingly complex digital ecosystems.